

SYSTEM EXECUTABLES INDEX

Developer: [Left Blank] | OptiCrop Project Documentation

Repository: <https://github.com/goutham-05-raj/OptiCrop-Smart-Agricultural-Production-Optimization-Engine>

1. Description of Source Executables

File Name	Location	Description
app.py	Root	Main Flask application server containing routes and logic.
train_model.py	Root	Runs data preprocessing, fits models, and saves pipeline artifacts.
generate_datasets.py	Root	Generates the mock datasets used to train models.
tests/test_app.py	tests	Automated test suite.
generate_datasets.py	tests	Synthetic dataset generator.
pytest.ini / tests	tests	Configuration and scripts for unit testing.

2. Output Executables

models/best_model.pkl	RF Classifier pipeline
models/yield_model.pkl	RF Regressor pipeline
models/scaler.pkl	StandardScaler
models/kmeans_model.pkl	K-Means (k=5)
models/yield_model.pkl	RF Regressor pipeline
models/scaler.pkl	StandardScaler
requirements.txt	Pip dependency list
README.md	Setup & run instructions

Configuration

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